

Reviewer Pack — Minimal Rank of Free Probability of Communication Channels

Entry 8662c185d7fe · INCONCLUSIVE

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Minimal Rank of Free Probability of Communication Channels

Entry ID: 8662c185d7fe **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-27 08:33:24 UTC

1. Conjecture

Field A (mathematical branch): Free Probability Theory **Field B** (complexity object): Communication Complexity

Statement:

{‘text’: ‘For any communication channel C with input and output alphabets having size n , the minimal rank of its free probability distribution $\rho(C)$ is upper bounded by a function $\omega(n)$ such that $\omega(n) = O(\log n)$.’, ‘quantitative’: ‘ $E[\text{rank}(\rho(C))] \leq \omega(n)$ ’, ‘falsifier’: ‘There exists a communication channel C with input and output alphabets having size n for which the rank of its free probability distribution $\rho(C)$ is greater than $\omega(n)$ ’}

Rationale (proposer’s reasoning):

{‘text’: ‘Free probability theory, by exploring non-commutative algebraic structures, may provide new insights into the complexity of communication processes. A connection between free probability and communication complexity could reveal hidden structures in the information flow, potentially leading to improved lower bounds or better algorithms.’, ‘invariant structure’: ‘The use of free probability as an invariant for communication channels aims to uncover algebraic properties that are not apparent with classical probability measures.’, ‘expected outcome’: ‘This bridge may expose non-trivial relationships between free probability and the complexity of information transmission, potentially leading to new insights in both fields.’}

Taxonomy category: BP_READTWICE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): b384ef579cc699c0

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The minimal rank of free probability distribution $\rho(C)$ for communication channels with input/output alphabet sizes n is supported if the mean rank across all generated channels $E[\text{rank}(\rho(C))] \leq \omega(n)$, where $\omega(n)$ is $O(\log n)$. Falsified if any channel has a rank greater than $\omega(n)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	1.00	UNCERTAIN	SAFE
KARP_LIPTON	SAFE	1.00	SAFE	SAFE

4. Novelty audit

Verdict: ADJACENT_OK against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - free probability AND communication complexity AND minimal rank - rank($\rho(C)$) $\leq \omega(n)$ AND communication channel - communication channels AND free probability theory AND $O(\log n)$

Top relevant hits considered: - [http://arxiv.org/abs/2410.20094v1] The Communication Complexity of Approximating Matrix Rank - [http://arxiv.org/abs/cs/0001014v3] Nondeterministic Quantum Query and Quantum Communication Complexities - [http://arxiv.org/abs/2501.05829v2] Satellite-based communication for phase-matching measurement-device-independent quantum key distribution - [http://arxiv.org/abs/2105.13085v3] Constraints on dark photon dark matter using data from LIGO's and Virgo's third observing run - [http://arxiv.org/abs/2206.14521v2] Reconstruction of interactions in the ProtoDUNE-SP detector with Pandora - [http://arxiv.org/abs/2510.14732v2] Measurement of CP asymmetry in $D^0 \rightarrow K_S^0 K_S^0$ decays with Run 3 data - [http://arxiv.org/abs/2009.01190v1] Properties and astrophysical implications of the 150 Msun binary black hole merger GW190521 - [http://arxiv.org/abs/2006.12611v1] GW190814: Gravitational Waves from the Coalescence of a 23 $M_\odot$ Black Hole with a 2.6 $M_\odot$ Compact Object

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, $\leq 90s$ wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_channel(n):
        X = ''.join(random.choices('abcdefghijklmnopqrstuvwxyz', k=n))
        Y = ''.join(random.choices('abcdefghijklmnopqrstuvwxyz', k=n))
        P = [[Fraction(random.random()) for _ in range(n)] for _ in range(n)]
        for row in P:
            total = sum(row)
            for i in range(n):
                row[i] /= total
        return X, Y, P

    def free_probability_distribution(P):
```

```

n = len(P)
I = [[Fraction(1 if i == j else 0) for j in range(n)] for i in range(n)]
A = [P]
B = [I]

while True:
    new_A = []
    new_B = []
    for a, b in zip(A, B):
        new_a = [[a[i][j] * b[k][l] + a[j][k] * b[l][i] for l in range(n)] for k in range(n)]
        new_b = [[b[i][j] * b[k][l] - a[i][k] * b[l][j] for l in range(n)] for k in range(n)]
        new_A.append(new_a)
        new_B.append(new_b)
    A.extend(new_A)
    B.extend(new_B)

    if len(A) > 2 * n:
        break

rank = sum(1 for a, b in zip(A, B) if any(a[i][j] != 0 or b[i][j] != 0 for i in range(n) for j in range(n)))
return rank

n_values = [5, 10, 15, 20, 30, 40]
ranks = []

for n in n_values:
    channel = generate_channel(n)
    rank = free_probability_distribution(channel)
    ranks.append(rank)

mean_rank = sum(ranks) / len(ranks)
conjecture_holds = all(rank <= math.log(n) for n, rank in zip(n_values, ranks))
counterexample = next((f"Rank {rank} > log({n})" for n, rank in zip(n_values, ranks) if rank > math.log(n)))

return {
    "metric_name": "rank",
    "metric_value": mean_rank,
    "instances_tested": len(ranks),
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys

    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_rank = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

```

```

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_rank} std=0.0 support_fraction={support_fraction}")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_rank} std=0.0 support_fraction={support_fraction}")
else:
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
    print(f"RESULT: FALSIFIED counterexample='{results[first_failing_seed]['counterexample']}'")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb3d0e97.py", line 81, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb3d0e97.py", line 59, in run_trial
    rank = free_probability_distribution(channel)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cb3d0e97.py", line 41, in free_probabil
    new_a = [[a[i][j] * b[k][l] + a[j][k] * b[l][i] for l in range(n)] for k in range(n)]
              ^

```

UnboundLocalError: cannot access local variable 'i' where it is not associated with a value

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from verifying the conjecture's support or falsification. | next: Investigate and fix the crash in the test code to proceed with the verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13739
2	propose	ollama_remote	glm4:latest	0	0	12597
3	preregistration	ollama_remote	glm4:latest	0	0	5614
4	novelty	ollama_remote	glm4:latest	0	0	4526
5	novelty	ollama_remote	glm4:latest	0	0	6302
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	26042

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10237
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9714
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10538
10	judge	ollama_remote	glm4:latest	0	0	10871

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 110178 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/8662c185d7fe.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/8662c185d7fe.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/8662c185d7fe.tar.gz (if generated)